

Mohanna Hoveyda

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I am a third-year PhD candidate at Radboud University, specializing in *Knowledge-Enriched Conversational AI*. My research interests encompass; 1) Employing neuro-symbolic solutions for **reasoning** with language models, and 2) Developing **adaptive** systems that dynamically choose the optimal answering strategy/agent—providing a direct response, retrieving information, or engaging in reasoning—based per user input characteristics.

EDUCATION

PhD, Radboud University, iCIS Nijmegen, Netherlands
Knowledge-Enriched Conversational AI February 2023
Advisors: Arjen P. de Vries, Faegheh Hasibi, Maarten de Rijke

MSc, University of Tehran, FNST Tehran, Iran
Computational Linguistics September 2020–January 2023

Bachelor, University of Tehran, ECE & FFL Tehran, Iran
Computer Engineering (Minor) 2018–2020
French Language and Literature (Major) 2016–2020

Highschool, National Organization for Development of Exceptional Talents (NODET) Karaj, Iran
Mathematics and Physics 2012–2016

RESEARCH

Adaptive Orchestration of Modular Generative Information Access Systems SIGIR 2025
Co-Authors: Harrie Oosterhuis, Arjen P. de Vries, Maarten de Rijke, Faegheh Hasibi

- Proposed a self-organizing GenIA architecture enabling real-time adaptive orchestration of LLM agents, retrieval models, and external tools per query to maximize relevance and minimize computational overhead.
- Developed a dynamic QA system that adaptively selects optimal answering strategies based on question complexity, using *contextual multi-armed bandits*.

Real-World Conversational Entity Linking Requires More Than Zero-Shots ACL 2024 Findings
Co-Authors: Arjen P. de Vries, Faegheh Hasibi, Maarten de Rijke

- Introduced benchmark and evaluation scenarios for more realistic assessment of entity linking models

Network Science-Based Structural Analysis of Semantic Representations Master's Thesis
Supervisors: Mostafa Salehi, Mahmood Bijankhan, Paulino Villas-Boas

- Applied network science metrics to analyze and compare machine-learned and human-curated semantic spaces across both uni-modal and multi-modal representations
- Developed a novel statistical method that enables comparing networks of varying sizes

EXPERIENCE

Teaching Assistant, Information Retrieval 2023 - 2024
Radboud University, Nijmegen, Netherlands

Teaching Assistant, Machine Learning for Natural Language Processing 2022
University of Tehran, Tehran, Iran

WORKSHOPS AND SCHOOLS

The European Summer School on Information Retrieval (ESSIR)
Amsterdam, Netherlands, University of Amsterdam July 2024
Vienna, Austria, Technische Universität Wien August 2023

High-Performance Computing (HPC) and Deep Learning Workshop August 2022
Tehran, Iran, HPC Laboratory, Institute for Research in Fundamental Sciences (IPM)

Summer School of Intelligent Learning*July 2019–August 2019*

Tehran, Iran, Institute for Research in Fundamental Sciences (IPM) and Sharif University of Technology

Autumn School of Mind and Brain*November 2018*

Tehran, Iran, Cognitive Sciences and Technologies Council and The University of Tehran

AWARDS AND SCHOLARSHIPS

Ranked 6th in Computational Linguistics National Master's Entrance Exam

August 2020

Iran National Organization for Educational Testing

Scholarship for Computational Linguistics Master's Program

August 2020

Tehran, Iran, University of Tehran

2nd Prize for The Best Actress in The 4th Student Festival of Francophone Theater

May 2019

Tehran, Iran, Iranian Association of French Literature and Language (AILLF)

Scholarship for Bachelor of French Literature

August 2016

Tehran, Iran, University of Tehran

SKILLS

Languages: Persian (Native), English (Fluent), French (Intermediate), Dutch (Basic)**Programming:** Python (Advanced), C (Intermediate), C++ (Intermediate), HTML (Basic), CSS (Basic)**Frameworks & Other Computing Skills:** Natural Language Processing (NLTK, SPACY, Gensim, Huggingface, Flair), Deep Learning (PyTorch, Tensorflow, Scikit-learn), Others (Networkx, Numpy, Seaborn), Praat**REFERENCES**

Arjen P. de Vriesa.devries@cs.ru.nl

Professor of Information Retrieval at Radboud University and Research Director of iCIS

Maarten de Rijkem.derijke@uva.nl

Professor of Artificial Intelligence and Information Retrieval at the University of Amsterdam

Harrie Oosterhuisharrie.oosterhuis@ru.nl

Assistant Professor at iCIS, Radboud University and Visiting Research Scholar at Google DeepMind